

A Packing of 12 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.423926. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.407101, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.423926 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 12$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.407101, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 12/11 = 2.424242$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N12.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
605981591612691968767104492645835685771444180732987748000725165858758491
694146879398405428460472897998920487827331485411436666624019610546718537
703750415080923953670817055124798144827456589273
```

denominator:

```
249999999999579603857741100848210058270631875925840196691168229250215500
208128421804626605848487986518974629881568766120053274242363589929920286
612469145878850131816752386297390004250641468620
```

B Coordinates

12 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N12.txt`.

```
-241885264026 -120058423382 211588332921 770658727573936496277 -32358596017 -163017322114 -4430042704237444167059 -642087126869 486145295273 -103996971245
-220204703699 373940947668 93316009505 -3909191660757812906906 205827591252 416997050454 -14142533595 -44394581600 334657339768 125055288136 -533540813311
274682217940 -631294768935 -365364771568 -1758672443574254685626 -168298983400 -290219618170 -19687363199571902522 271962581276 238385970032 378736921815
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-76381748359 83987019569 237911495019 -175456136273122994552765 -342138621806 509223290899 1567717566851576235149 187906264454 45433055377 651351978059
-510823273946 260931655189 -234750993377 2475496968800621161789 591443137017 -24603067537 58191495170528428766035 -236043330603 279031422187 215444798805
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-356633631585 -257287317628 180075999325 -211907680488288683455 325827191049 341098601462 -97928454307-32928890141 -147921476312 -264256834291 326451053651
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```